



MICRO PRECISION CALIBRATION, INC.
22835 Industrial Pl
Grass Valley California 95949
5302681860

Certificate of Calibration

Date: August 05, 2025

Cert No. 5523631032252175

Customer:

FUERZA AEREA DE CHILE
RODRIGO DE ARAYA # 1263
SANTIAGO CHILE 7810161

MPC Control #: FA2644
Asset ID: N/A
Gage Type: VIBRATION TRANSDUCER
Manufacturer: CEC
Model Number: 4-123-0001
Size: 45 to 2000 Hz
Temp/RH: 22.9°C / 45 %
Location: Calibration performed at MPC facility

Work Order #: SD-50054553
Purchase Order #: CC
Serial Number: 31988
Department: N/A
Performed By: PADUNG SRASUAY
Received Condition: IN TOLERANCE
Returned Condition: IN TOLERANCE
Cal. Date: August 05, 2025
Cal. Interval: 12 MONTHS
Cal. Due Date: August 05, 2026

Calibration Notes:

See attached calibration data. (1 page)

This report replaces obsolete report # 5523631032050751.
Company name changed per customer request.

Standards Used to Calibrate Equipment

I.D.	Description.	Model	Serial	Manufacturer	Cal. Due Date	Traceability #
DX4383	DIGITAL MULTIMETER	34411A	SG48000118	AGILENT	Apr 30, 2026	5523631031792040
DX4378	DIGITAL MULTIMETER	34411A	MY48005467	AGILENT	Apr 30, 2026	5523631031792164
DM8728	PRECISION CONDITIONING AMPLIFIER	2650	1923018	BRUEL & KJAER	Jul 31, 2026	5523631032011707
DZ9999	REFERENCE STANDARD ACCELEROMETER	8305-001	1450416	BRUEL & KJAER	Nov 30, 2025	A2.00-5578-0.1

Calibrating Technician:

PADUNG SRASUAY

QC Approval:

JACK WERTZ III

STATEMENTS OF PASS OR FAIL CONFORMANCE: The uncertainty of measurement has been taken into account when determining compliance with specification. All measurements and test results guard banded to ensure the probability of false-accept does not exceed 2% in compliance with ANSI/NCSL Z540.3-2008.

THE CALIBRATION REPORT STATUS:

PASS - Term used when compliance statement is given, and the measurement result is PASS.

PASS² - Term used when compliance statement is given, and the measurement result is conditional passed or PASS².

FAIL - Term used when compliance statement is given, and the measurement result is FAIL.

FAIL² - Term used when compliance statement is given, and the measurement result is conditional failed or FAIL².

REPORT OF VALUE - Term used when reported measurement is not requiring compliance statement in report.

ADJUSTED - When adjustments are made to an instrument which changes the value of measurement from what was measured as found to new value as left.

LIMITED - When an instrument fails calibration but is still functional in a limited manner.

The expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor $k=2$, which for a normal distribution corresponds to a coverage probability of approximately 95%, unless otherwise stated. This calibration report complies with ISO/IEC 17025:2017, ANSI/NCSL Z540.3-2008 and ANSI/NCSL Z540.1-1994. Calibration cycles and resulting due dates were submitted/approved by the customer. Any number of factors may cause an instrument to drift out of tolerance before the next scheduled calibration. Recalibration cycles should be based on frequency of use, environmental conditions and customer's established systematic accuracy. All standards are traceable to SI through the National Institute of Standards and Technology (NIST) and/or recognized national or international standards laboratories. Services rendered include proper manufacturer's service instruction and are warranted for no less than thirty (30) days. The information on this report pertains only to the instrument identified; this may not be reproduced in part or in a whole without the prior written approval of the issuing MP Calibration Laboratory.



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Procedures Used in this Event

Procedure Name	Description
MPC-VIB-001 Rev. 06	Accelerometers, General, Rev.06, Jun-07-2024

Calibrating Technician:

S. Padung
PADUNG SRASUAY

QC Approval:

Jack R. Wertz III
JACK WERTZ III

STATEMENTS OF PASS OR FAIL CONFORMANCE: The uncertainty of measurement has been taken into account when determining compliance with specification. All measurements and test results guard banded to ensure the probability of false-accept does not exceed 2% in compliance with ANSI/NCSL Z540.3-2006.

THE CALIBRATION REPORT STATUS:

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FAIL - Term used when compliance statement is given, and the measurement result is FAIL.

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Calibration Report of CEC 4-123-0001 Vibration Transducer

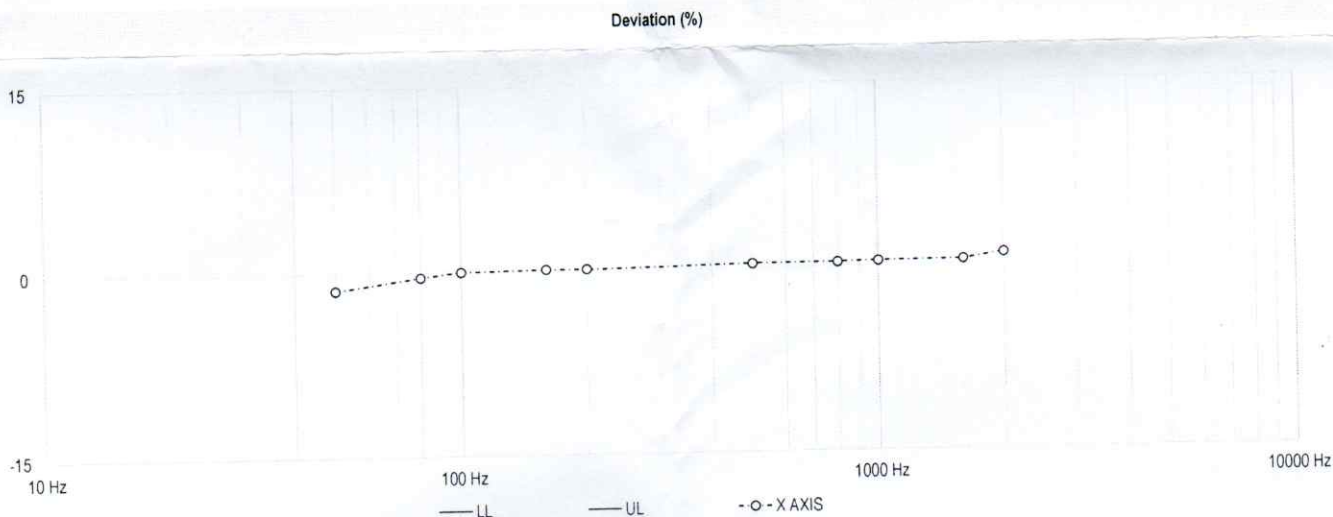
MPC Control #: <u>FA2644</u>	Serial Number: <u>31988</u>
Asset ID: <u>None</u>	Calibration Date: <u>August 05, 2025</u>

Reference Sensitivity

Applied	UUT Spec Sensitivity	Measured Sensitivity	Deviation
		mV/in/s	
100 Hz	135.000 mV/in/s	139.531	3.36 %

Frequency Response

Applied	UUT Ref Sensitivity	Measured Sensitivity	Deviation
		mV/in/s	
50 Hz	139.531 mV/in/s	137.592	-1.39 %
80 Hz	139.531 mV/in/s	138.975	-0.40 %
100 Hz	139.531 mV/in/s	139.531	0.00 %
160 Hz	139.531 mV/in/s	139.687	0.11 %
200 Hz	139.531 mV/in/s	139.699	0.12 %
500 Hz	139.531 mV/in/s	139.987	0.33 %
800 Hz	139.531 mV/in/s	140.028	0.36 %
1000 Hz	139.531 mV/in/s	140.097	0.41 %
1600 Hz	139.531 mV/in/s	140.159	0.45 %
2000 Hz	139.531 mV/in/s	140.865	0.96 %



The user shall determine the suitability of the equipment for its intended use. The calibration status is defined as **Report of Value**.

- End of Calibration Report -